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Artificial intelligence and autonomy at work: empirical insights from Germany

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Abstract
Artificial intelligence (AI) is a prominent topic regarding the digitalisation of work and its diffusion is expected to radically change job quality. Overall, there exists a large discrepancy between discursive expectations and quantitative empirical evidence. In this article, we use a novel module from the German Socio-Economic Panel to examine the overall prevalence of AI at work, the determinants that increase the likelihood of AI use, and its association with autonomy. The results show that 38% of German workers use AI, and AI use is associated with the use of specific digital technologies. Workers in high-level, non-routine occupations are more likely to use AI, particularly in comparison to manual workers. Moreover, the association between AI and autonomy is merely superficial and cannot be properly evaluated without considering workplace preconditions.

Keywords Artificial intelligence, Digital technology, Autonomy, Work, Job quality, Occupations

JEL Classification J00, J24, O3, O33

1 Introduction

Artificial intelligence (AI) dominates political and scientific debates on the digitalisation of work (Spencer 2018; Warhurst and Hunt 2019), and given the latest developments in text-generating AI and its diffusion (Green et al. 2023; Warning 2023), AI is portrayed as a central and disruptive technology that will cause radical changes in the working world (OECD 2019; European Commission 2020). Although these predictions are not new many insist that 'this time it is different' (Wajcman 2017; Boyd and Holton 2018). Highly noted publications have focused on the potential role of AI in job substitution (Frey and Osborne 2013; Benzell et al. 2015; Ford 2015; Webb 2020) or task substitution (Autor 2015; Arntz et al. 2016; Dengler and Matthes 2018; Fregin et al. 2023; Grienberger et al. 2024). Moreover, researchers have suggested that only workers of specific occupational levels use AI to their advantage, leading to major transformations in their work (Moore 2019; Lane and Saint-Martin 2021; Konle-Seidl and Danesi 2022). Additionally, AI is often assumed to have a definite impact on job quality (Spencer 2018; Lane and Saint-Martin 2021; Green et al. 2023; Lane et al. 2023). Authors argue that AI can either increase worker autonomy (Goos et al. 2019; Lane and Saint-Martin 2021) or lead to a loss of autonomy (Gerst 2019; Moore 2019; Milanez 2023), primarily through task substitution or complementary effects.

Despite the impactful research on AI potential, most discussions on AI and work, however, lack empirical quantitative evidence (Giering 2022; Merkel-Kiss and von Garrel 2022; Vivarelli and Arenas Díaz 2025). The discourse rarely considers AI as part of the long-term digitalisation of work, which is gradually being integrated

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into workplaces in association with specific technological preconditions. Consequently, studies that reflect the current status of firms (Holm and Lorenz 2022) and the complexity and uncertainty of technological diffusion remain rare (Liker et al. 1999; Orlikowski 2007; Hirsch-Kreinsen 2016). Research exploring how the trajectories of AI and work are shaped by human agency, as well as by existing social, technical, and economic conditions of work, is scarce (Howcroft and Taylor 2014, 2023; Hirsch-Kreinsen 2016). Thus, in summary, it is currently empirically unknown which occupational levels use AI at work and what other determinants (e.g., workers' education) are associated with AI use. Moreover, although the current discourse assumes that AI has a definitive impact on job quality, it remains unclear whether such (if any) associations between AI use and autonomy can be demonstrated empirically.

To address these research gaps, this article examines (1) which determinants are empirically associated with AI use, and (2) whether it is possible to identify empirical associations between AI use and autonomy beyond the existing associations with other digital technologies that are already used at work. Therefore, a novel survey module from the representative German Socio-Economic Panel (SOEP) is used. First introduced in 2020—prior to the emergence of widely used generative AI applications—this module includes questions on AI-based systems, digitalised workplaces, and job quality, while covering a representative sample of the adult German workforce. Germany serves as a particularly compelling case, given its status as one of the world's leading economic actors. Therefore, it is often portrayed at the forefront and as an example of digital transformation (Arntz et al. 2019; Marx et al. 2022b). This article contributes to the literature by providing quantitative empirical evidence of the overall prevalence and determinants of AI use at work. Additionally, it expands knowledge on the associations between AI use and job quality, particularly regarding job autonomy, while acknowledging technical preconditions of workplaces, workers' education, and their occupational level.

2 Literature review

2.1 AI use, occupational level, and technological preconditions in the workplace

Publications on AI and work often assume that specific occupational levels use AI, leading to significant changes in the working world (Arntz et al. 2019; Moore 2019; Lane and Saint-Martin 2021). The main question is whether the digitalisation of work would upgrade or polarise workers' skills and tasks (Hirsch-Kreinsen 2016). An optimistic view predicts an upgrade in skills and tasks at all occupational levels through workers' digital empowerment, which results in increased information availability, flexibility, and labour productivity (Brynjolfsson and McAfee 2012; Evangelista et al. 2014). Following this perspective, a broadly equal use of AI across all occupational levels may be found, offering benefits to all workers.

Meanwhile, the 'skill-biased technical change' perspective indicates that the diffusion of digital technologies will benefit occupational levels who already have higher skills and autonomy, owing to the increased automation of low-skilled and routine tasks (Autor et al. 2003). Similarly, studies have highlighted that the use of digital technologies may polarise jobs and skills, as the tasks of medium-skilled workers can be substituted more easily by AI than those of low- and high-skilled workers (Autor and Dorn 2013; Brynjolfsson and McAfee 2014; Goos et al. 2019). Occupational levels involving complex cognitive tasks will become more important, while concurrently, new, low-skilled, simple tasks will emerge that cannot (yet) be automated due to technological limitations, often referred to as 'engineering bottlenecks' (Frey and Osborne 2013). This perspective on polarisation suggests that AI use should prevail at high-skilled occupational levels as a substitute for low-skilled, routine tasks to allow focus on more complex tasks. Moreover, recent studies have argued for the complementary use of AI in medium-skilled occupations, comprising a combination of routine and non-routine tasks, each performed by machines and humans according to their unique strengths (Autor 2015). Therefore, AI use may be found in medium-skilled occupations.

However, complex software such as AI-based systems must be embedded in technical infrastructure comprising hardware and software layers (Rammert et al. 1998; Yoo et al. 2010; Ametowobla 2022), in the form of digital technologies, such as personal computers (PCs), smartphones, or enterprise resource planning (ERP) systems. Although this is evident, it is often neglected in current debates. In workplaces where certain digital technologies are used frequently, AI use is also likely to be more widespread. Given the uneven diffusion of digital technologies across occupational and educational levels (Friedrich et al. 2021), the idea of AI diffusion based on the technical preconditions of the workplace and occupational level of workers is supported.

Existing empirical research indicates that between 11% (Pfeiffer 2022) to 25% (Meyer et al. 2022) of German workers use AI. While this research already provides valuable first insights into the diffusion of AI in Germany, representative data for the entire German workforce is still lacking. Evidence from Denmark shows that AI is predominantly used at work by those in medium-skilled occupations (Holm and Lorenz 2022), whereas findings from Germany indicate that AI is used mostly in high-level occupations (Meyer et al. 2022). Furthermore, AI

appears to increase the need for highly educated workers (Lane et al. 2023).

In conclusion, empirical evidence on AI use in the workplace remains limited; theoretical arguments suggest that AI is either used as a technological advantage across all occupational levels or used predominantly in high-level occupations, where low-skilled, routine tasks can be substituted by AI-based systems. However, it is still unclear which other determinants are associated with AI use at work. Hence, this article examines the associations between AI use and workers' occupational levels and technical preconditions (in form of digital technologies) in the workplace to gain more differentiated insights into the use of AI at work.

2.2 AI and autonomy in the workplace

A crucial aspect of the debate on AI in the workplace is the assumption that it affects job quality (Green et al. 2023). Autonomy, a key aspect of job quality (Kalleberg 2011; Findlay et al. 2013; OECD 2013), is central to this debate (Kirchner et al. 2023). The prevailing notion is that AI has the potential to replace specific tasks, thus affecting worker autonomy, which could increase when repetitive and routine tasks are performed by AI. Therefore, workers have more freedom to handle complex, nonroutine cognitive tasks that machines cannot handle (Goos et al. 2019; Lane and Saint-Martin 2021) such as those requiring abstract thinking, creativity, and empathy (Moore 2019; Wilkens 2020). This viewpoint seems reasonable because highly complex activities that require contextual, social, or relational understanding cannot be performed by AI yet (Pettersen 2019). However, AI can support the handling of data and predictive knowledge (Hirsch-Kreinsen 2016; Ramge 2018) while workers ultimately decide how to fulfil a given task. This increases worker autonomy because workers would rely more on their individual skills to complete specific tasks. Thus, AI use would be systematically associated with increased autonomy.

In contrast, others suggest that workers may enjoy less autonomy if they are merely expected to follow automated instructions from machines (Gerst 2019). Specifically, AI can worsen the quality of work because of its monitoring and tracking capabilities (Spencer 2018), resulting in micromanagement and reduced worker autonomy (Moore 2019). As simple routine tasks are substituted by machines, workers are left with only non-automatable and tedious tasks (Green et al. 2023). More demanding or challenging tasks and time savings that must be used efficiently can increase the pace and complexity of work, leading to a decline in job quality (Moore 2019; Milanez 2023). As 'engineering bottlenecks' decrease (Dengler and Matthes 2021), not only manual tasks may be substituted, but also complex, cognitive work (Heinen et al. 2017), resulting in the devaluation of human experiential knowledge (German Federal Ministry of Labour and Social Affairs 2017). Overall, this perspective suggests that AI use is systematically associated with reduced autonomy.

Empirically, the impact of AI on job quality and autonomy remains ambiguous (Lane and Saint-Martin 2021), and most evidence is based on case studies. Yamamoto (2019) reported that AI use has led to the restructuring of tasks from routine to more complex non-routine tasks while increasing worker satisfaction and work-related stress. Similarly, according to Lane et al. (2023), workers reported improved work performance, and therefore, higher job quality as a result of using AI. Yet, three-quarters of AI users reported that AI increased the pace of work, leading to increased work intensity. Milanez (2023) showed that the use of AI improved the overall job quality because it reduced the time spent on simple, tedious tasks and left workers with more freedom for complex tasks, which led to a reduction in tedium and better work engagement.

However, it must be emphasised that AI-based systems are embedded in digital technologies, such as PCs and ERP systems, that are already used at work, and these technologies have been associated with autonomy. For instance, studies have found that the use of digital technologies tends to be associated with higher autonomy (Chesley 2014; Kirchner 2015; Meyer and Hünefeld 2021), depending on the task (Kirchner et al. 2023) and technology (Meyer et al. 2019). By contrast, digital work instructions and monitoring are associated with less autonomy (Gensler and Abendroth 2021; Marx et al. 2022a). Moreover, the relationship between autonomy and digital technology seems paradoxical (Mazmanian et al. 2013; Ruiner and Klumpp 2020), simultaneously promoting autonomy and monitoring (Gerten et al. 2019), whereas workers strategically avoid digital instructions to enhance their autonomy (Terry et al. 2022).

The overall perspective on embedding AI in existing digital technologies suggests an overlap between autonomy, digital technologies, and AI use. If analyses focus only on AI use and autonomy, we may find positive or negative associations, as theoretically suggested. Thus, an empirical examination of this fundamental association is necessary to guide the research field. However, whether these associations are truly related to the use of AI remains unclear. AI use is likely associated with digital technologies used at work, which, in turn, are associated with autonomy. Following this argument, we assume that we can estimate the relationship between AI use and autonomy only by considering its interplay with technical preconditions in the workplace, specifically in the form of digital technologies.

3 Data and methods

3.1 Data

The analyses in this article were based on cross-sectional data from the 37th wave of the SOEP in 2020, a representative annual panel study of private households, including approximately 30,000 individuals from 19,000 households in Germany. The panel included extensive socioeconomic characteristics along with questions on satisfaction, employment, education, and health (Goebel et al. 2019). In 2020, a novel questionnaire module on the digital transformation of the working world was first fielded (Fedorets et al. 2021). The module included questions on AI-based systems, digital workplaces, and job quality. Given that the research focus is exclusively on employed individuals, non-employed respondents were excluded from the sample (Table A1 in the appendix). Consequently, the sample consisted solely of working adults, including workers, employees, civil servants, and the self-employed, who provided information about AI usage. After excluding cases with missing information in the regression analyses, the final samples comprised 11,649 (Table 3) and 13,511 respondents (Table 4).

3.2 Variables

3.2.1 Dependent variables

'AI use' was the dependent variable in the first regression analysis (see Table 3), which analysed the determinants of AI use at work. The underlying question of this variable was, 'Nowadays, some of the tasks performed in the workplace can be done by digital systems. In your job, how often do you work with digital systems that…?'. Five different tasks were given: automatically recognize and process: (1) 'language or spoken commands'; (2) 'images, videos, and photos'; (3) 'text, handwriting, or numbers'; (4) 'information or data and independently evaluate this information'; and (5) 'questions requiring specialised knowledge'. The respondents answered every item on a five-point Likert scale ranging from 'never' (5) to 'several times a day' (1). All scales were reversed, such that higher numerical values corresponded to higher frequencies (Table A1 in the appendix presents descriptive statistics for these items). Based on these items, a binary variable ('AI use') was generated to address the first research question regarding the determinants of AI use at work, thereby capturing whether respondents reported using at least one of the five AI-based systems or no AI-based system at all (1 = yes, 0 = no; sample mean = 0.38; SD = 0.48). A binary variable was used because the frequency distributions of the five items were skewed (Fig. 1); thus, most of the variance was between the two poles of 'never' and '(any) use'. Hence, minimal information is lost when a binary variable is used.

A fundamental problem in AI research is the lack of a universally accepted definition for the broad (technological) concepts of AI (OECD 2019). AI is generally described as a field of computer science that aims to enable machines to perform narrowly defined tasks intelligently (Moore 2019), without necessarily providing a detailed definition of what 'intelligence' entails or specifying the technologies involved. The question of digital systems performing specific tasks autonomously, as examined in this empirical analysis, aligns with the definition of narrow or weak AI. This type of AI is primarily used for machine learning-based pattern recognition in unstructured large data sources (such as texts, images, or datasets) to make predictions or provide recommendations. Therefore, it captures the currently most common application of AI (Döbel et al. 2018; Konle-Seidl and Danesi 2022). Notably, it was assumed that these AI-based systems were sufficiently diffused to be queried in a general household survey, and in contrast to existing research, the methodological approach avoided the unspecific term 'AI', and asked indirectly about AI at work. This is because workers may not be aware that they are using AI-based systems (Giering et al. 2021), and the term is still opaque and discursively loaded (Holton and Boyd 2021; Overdiek and Petersen 2022; Brauner et al. 2023). However, the respondents should be able to assess whether they use certain digital systems for specific tasks at work.

The level of autonomy at work ('job autonomy') was the dependent variable in the second regression analysis (Table 3), which analysed the association between autonomy and AI use.[1] Following the classic literature on job autonomy (Kalleberg 2011; Muñoz de Bustillo et al. 2011; Eurofound 2015), the operationalisation was based on five items: 'If you think about your situation at work in the last month, how often did you (1) decide by yourself how to do your work, (2) set the pace of your work by yourself, (3) determine the order in which you completed your tasks, (4) learn new things at work, and (5) solve unanticipated problems independently while working?'. Respondents answered on a five-point Likert scale ranging from 'never' (5) to 'always' (1). All scales were reversed, such that higher numerical values corresponded to higher frequencies (Table A1 in the appendix presents descriptive statistics for these items). Based on these five items, a

[1] The autonomy variable requires critical reflection, as it is not a fixed characteristic of certain occupations, but rather a dynamic factor of job quality that is shaped by social and institutional influences, as well as individual and collective practices (Wajcman 2006). It can change over time and have paradoxical effects on workers: on the one hand, technological devices can provide flexibility in performing work, but on the other hand, they can increase workers' involvement, thereby limiting their ability to disconnect from work (Mazmanian et al. 2013). Furthermore, high autonomy at work is not inherently positive and for some, too much autonomy can result in negative job quality, for instance, because of excessive or overwhelming demands. Accordingly, it will be essential for future research to explore in more detail how increased autonomy and the use of AI affect job quality, for example by drawing on indicators such as job satisfaction or occupational health.

mean index of job autonomy was generated. The sample mean for this index was 3.82 (SD = 0.90) and Cronbach's alpha was 0.83 for this five-item scale.

3.2.2 Independent variables

The questionnaire included questions on the digital technologies used by workers, covering more common technologies, such as PCs or laptops, and modern tools, such as robots and messaging programs. These digital technologies act as technical infrastructures, consisting of hardware and software layers (Rammert et al. 1998; Yoo et al. 2010; Ametowobla 2022), within which complex AI-based systems are embedded. For instance, AI-based scanning software could summarize and categorize the content of entire texts within seconds. This would not only enhance the use of AI-based systems (within scanners) but could also potentially affect workers' autonomy, as eliminating such time-consuming tasks would enable them to focus on complex and demanding activities. Respondents answered using the same five-point Likert scale for each digital technology and these scales were similarly inverted (Table A1 in the appendix presents descriptive statistics for these items). In addition to digital technologies at work, workers were asked what kind of education or training is usually required for their main occupations, as AI may increase the need for highly educated workers (Lane et al. 2023). Additionally, workers were asked about their current occupational levels based on the standard classification of Erikson et al. (1979; Ganzeboom and Treiman 2003).

3.2.3 Control variables

Furthermore, several control variables that could potentially bias the analyses were considered, including respondents' sociodemographic factors, such as age and gender (0 = female, 1 = male), as well as control variables related to the organisational level of the respondents' workplace. These included whether the workplace belonged to the IT/fintech sector (0 = no, 1 = yes), whether the employer was in the public sector (0 = no, 1 = yes), and company size (1–19, 20–199, and 200 or more people). These aspects are relevant to debates on the diffusion of digital technology and AI (Friedrich et al. 2021; Ohlert et al. 2022; Green et al. 2023).[2]

3.3 Methods

The binary 'AI use' variable was employed as the dependent variable in three logistic regression models (Table 3), which analysed the determinants of AI use at work. As the dependent variable was dichotomous, logistic regressions were performed (Kohler and Kreuter 2016).[3] Given that interpreting the coefficients of binary response models might be difficult (Mood 2009), the average marginal effects (AME) were presented to ensure an intuitive interpretation and comparison of the estimated effects across groups, samples, and models. The analyses followed a stepwise model approach to obtain detailed information regarding these associations.

For the second question, which focused on the association between job autonomy and AI use, the generated 'job autonomy' variable was used as the dependent variable in the ordinary least squares (OLS) regression models to examine the association between job autonomy and AI use. Similarly, a stepwise model approach was adopted to analyse in detail whether the associations between job autonomy and AI use extend beyond potentially existing associations.

To detect potential multicollinearity among the independent variables, which could bias the results, variance inflation factor (VIF) tests were conducted (see appendix for the results of these tests). To ensure the comparability of the results, for all regressions, only cases that did not have any missing values in the complete model were considered.

4 Results

4.1 Descriptive analysis

To visualise the number of workers using AI-based systems, corresponding weighted[4] bar charts were created.

Figure 1 illustrates that most workers across all five AI-based systems stated that they had 'never' used the respective AI-based systems at work (a 72–85% range), whereas the cumulative usage categories demonstrated the use of AI-based systems in a 15–28% range (Table 1). Cumulating the frequency categories indicating usage seems reasonable, as it is not possible to qualitatively evaluate the respective tasks of respondents' individual work based on the available data. Therefore, the rare use of a single AI-based system can be significant for individual workers.

Almost 28% of the workers used AI to recognise, process, and interpret information or data, whereas 21%

[2] Additional robustness checks for all models included information on the month of the interview, as well as subgroup analyses to control for potential effects of the COVID-19 pandemic. Descriptive analysis regarding the use of AI-based systems before and after the nationwide COVID-19 lockdown in Germany (end of march 2020), as well as models including the month of the interview, did not reveal any systematic differences in the relationships of interest that would concern the generalizability of the results.

[3] Robustness checks with a mean index variable of the five AI items (sample mean 1.43 [SD = 0.74], Cronbach's alpha = 0.82), and OLS regressions produced similar results.

[4] Weighted percentages were used in this context to compensate for sample biases, such as overrepresentation or underrepresentation of certain groups, for example, highly educated individuals. The SOEP provides the personal weighting variable 'phrf' for this purpose.

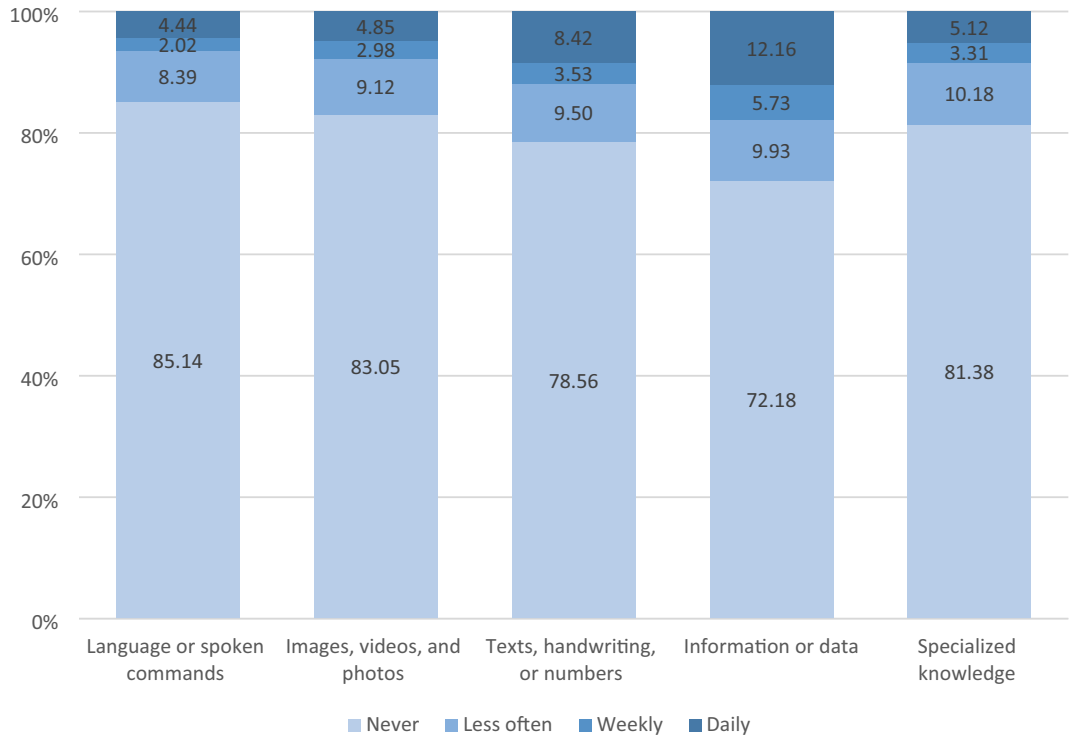

Fig. 1 Use of AI-based systems at work to recognise and process (weighted percentages). Source: SOEP v37 2020 (EU version); authors' own calculations

Table 1. Use of AI-based systems at work (weighted, cumulative percentages)

Digital systems to recognise and process…	Language or spoken commands	Images, videos, and photos	Texts, handwriting, or numbers	Information or data	Specialised knowledge	At least one AI-based system
Use	14.86	16.95	21.44	27.82	18.62	37.81
No use	85.14	83.05	78.56	72.18	81.38	62.19

Source: SOEP v37 2020 (EU version), authors' own calculations

used AI-based systems for text recognition and processing. Overall, approximately 38% of the workers in Germany used at least one AI-based system, whereas 62% never used any. The methodological approach of an indirect AI inquiry employed in the SOEP provides a more realistic assessment of the use of AI-based systems in the workplace, as it avoids direct reference to the ambiguous term 'AI,' which could bias the results. Consequently, this approach results in discrepancies, such as higher reported usage rates, when compared to previous surveys on AI usage in workplaces in Germany (Meyer et al. 2022; Pfeiffer 2022).

Table 2 shows the descriptive sample statistics for the variables by AI use. Men reported using AI-based systems more frequently than women, and younger workers tended to use AI-based systems more frequently than older workers. Workers using specific digital technologies also reported a higher frequency of AI use. This was particularly true for the respondents who used robots and ERP systems. People in occupations requiring higher education (technical and university degrees) reported using AI more frequently than those in occupations requiring less education. Descriptive sample statistics also demonstrated differences in occupational levels. Specifically, 56% of higher-grade managers and professionals reported using AI compared to only 18% of unskilled and semi-skilled workers. The otherwise self-employed also reported an above-average use of approximately 53%.

4.2 Multivariate analyses

Table 3 presents the results of the logistic regression models in which AI use was employed as a binary dependent variable to investigate whether the associations between AI use and the explanatory variables remained substantial while accounting for control variables.

Table 2. Descriptive sample statistics by AI use

Variable	*N* (unweighted)	Use of AI (weighted)
Dummy: Use of any AI-based system	17,006	37.81%
Sex: female	8,389	32.88%
Sex: male	8,613	42.25%
Age: younger than 35 years	4,467	41.71%
Age: 35–49 years	5,758	40.26%
Age: 50 years and older	6,780	33.18%
Workers using the following digital work tools		
Stationary PC, work station, server	11,680	46.58%
Laptop or notebook	8,800	51.5%
Smartphone or tablet computer	9,407	49.11%
Robot	1,022	69.69%
Scanner	9,651	50.46%
Digital measurement or diagnostic devices	3,835	55.11%
Programmes/apps that send/receive messages	11,685	48.65%
Programmes/apps that specify the times, places, or resources of work (ERP systems)	6,574	59.79%
Required education/training for respondents' occupation		
No vocational training	2,788	23.43%
Instruction/training/courses	702	14.23%
Completed vocational training	7,071	38.39%
Technical college degree	1,611	49.99%
University/other institution of higher education degree	2,676	50.94%
Occupational level		
High-grade manager/professional	2,997	56.56%
Low-grade manager/professional	4,058	45.49%
Routine non-manual employee, high-grade (in administration/commerce)	2,364	36.68%
Routine non-manual employee, low-grade (sales and services)	1,735	26.24%
Small self-employed with/without employees	499	29.24%
Skilled manual worker	1,302	28.58%
Semi- and unskilled manual worker	2,320	18.40%
Agricultural labourer	177	15.84%
Otherwise self-employed	582	53.50%

Source: SOEP v37 2020 (EU version); authors' own calculations

Model 1 shows a significant ($p < 0.001$) positive association between AI use and digital technologies at work. Specifically, the use of robots (a 4.1%-point higher probability) and ERP systems (a 4.5%-point higher probability) was most associated with AI use at work. Therefore, workers who used these two digital technologies were significantly more likely to use AI. The use of smartphones and tablets (a 1.0%-point higher probability) and digital measurement or diagnostic devices (a 2.2%-point higher probability) showed the weakest positive association. Model 1 highlights that AI use is significantly associated with certain digital technologies, which make it more likely to be used at work. While this may seem unsurprising at first glance, it has important implications regarding the impact of AI, as digital technologies, and therefore AI use, is not evenly diffused (see appendix). Workplaces where specific digital technologies are used are thus more likely to utilize AI-based systems. Therefore, a uniform use of AI across all workplaces seems unlikely.

Model 2 presents the findings on the associations between AI use and the training or education required in respondents' occupations, as well as their occupational levels. This model showed that compared with the reference group of occupations that did not require vocational training, workers who needed instruction or training for their occupation were less likely to use AI ($p < 0.05$). Workers in occupations that required a higher level of education or training (at least vocational training) were significantly more likely to use AI at work ($p < 0.001$). Workers whose occupations required a technical college (university) degree had a 12.9 (12.6) percentage-point higher probability of using AI at work than those in occupations requiring less education or training. Similarly, associations with the respondents' occupational levels revealed that compared with the reference group of lower-grade managers or professionals, only higher-grade managers or professionals had a significantly higher probability of using AI at work ($p < 0.001$). By contrast, all other occupational levels had a lower probability of

Table 3. Results of the logistic regressions

	(1) AI use	(2) AI use	(3) AI use
Controls	Yes	Yes	Yes
Digital technology			
Stationary PC, work station, server	0.027***		0.022***
	(0.003)		(0.003)
Laptop/notebook	0.035***		0.026***
	(0.003)		(0.003)
Smartphone/tablet	0.010***		0.010***
	(0.003)		(0.003)
Robot	0.041***		0.044***
	(0.007)		(0.007)
Scanner	0.024***		0.024***
	(0.003)		(0.003)
Digital measurement/diagnostic devices	0.022***		0.023***
	(0.003)		(0.004)
Programmes that send/receive messages	0.033***		0.026***
	(0.003)		(0.003)
Programmes that specify the times, places, resources of work (ERP systems)	0.045***		0.045***
	(0.003)		(0.003)
Education/training			
(ref. No completed vocational training)		0.000	0.000
		(.)	(.)
Instruction/training		−0.046*	−0.023
		(0.023)	(0.025)
Completed vocational training		0.086***	0.018
		(0.013)	(0.013)
Technical college degree		0.129***	0.042*
		(0.019)	(0.018)
University/institution of higher education degree		0.126***	0.030
		(0.018)	(0.017)
Occupational level			
(ref. Low-grade manager/professional)		0.000	0.000
		(.)	(.)
High-grade manager/professional		0.082***	0.047***
		(0.015)	(0.013)
Routine non-manual employee, high-grade (in administration/commerce)		−0.058***	−0.002
		(0.015)	(0.013)
Routine non-manual employee, low-grade (sales and services)		−0.153***	−0.044**
		(0.017)	(0.016)
Small self-employed with/without employees		0.030	−0.014
		(0.036)	(0.031)
Skilled manual worker		−0.180***	−0.067***
		(0.017)	(0.018)
Semi- and unskilled manual worker		−0.224***	−0.062***
		(0.016)	(0.018)
Agricultural labourer		−0.253***	−0.117*
		(0.036)	(0.045)
Otherwise self-employed		0.014	0.081
		(0.243)	(0.234)
Pseudo R^2	0.1724	0.0905	0.1784
N	11,649	11,649	11,649

Note: The average marginal effects (AME) are shown. The standard errors are indicated in parentheses. The number of cases was fixed for Model 3 to avoid differences owing to missing values. The full results with the logit regression coefficients and controls are provided in the appendix

Source: SOEP v37, 2020 (EU version); authors' own calculations. For significance levels, * denotes $p < 0.05$; **, $p < 0.01$; and ***, $p < 0.001$

using AI ($p < 0.001$). The effects were particularly strong for unskilled/semi-skilled workers (a −22.4%-point lower probability of use), agricultural labourers (a −25.3%-point lower probability of use), and lower-grade routine non-manual workers (a −15.3%-point lower probability of use). No statistical association was found between AI use by individuals running small self-employed businesses and those otherwise self-employed. Therefore, Model 2 indicates that respondents in occupations requiring a higher level of education or training and those in high occupational levels are more likely to use AI at work.

Model 3 included all the determinants. The positive association with digital technology remained robust and significant in all cases ($p < 0.001$), even when the required levels of education or training and occupational level were added. In this case, the AMEs of the digital technologies changed minimally, if at all, indicating their orthogonality to the other variables in the model. This is an important finding, as theoretical assumptions suggested that the occupational level, in particular, influences AI use. However, we observe that digital technologies play a crucial role in the usage of AI-based systems at work. Regarding the education or training required for respondents' occupations, the effects and their significance changed considerably compared to those in Model 2. Although no significant associations with AI use was found for workers in occupations that required instruction or training, vocational training and university degree, the effect size and significance level of the technical college degree group decreased but remained unchanged in its statement. Thus, similar to Model 2, Model 3 shows that workers in occupations requiring a higher education (technical degree) were still more likely to use AI than those in occupations that did not requiring vocational training. A similar picture emerged regarding respondents' occupational levels, wherein the effects and significance of the individual levels decreased considerably. Nonetheless, higher-grade managers or professionals were still more likely to use AI than the reference group of lower-grade managers or professionals, albeit with a reduced effect size. Similarly, skilled and unskilled or semi-skilled manual workers, agricultural labourers, and lower-grade routine non-manual employees were less likely to use AI at work than lower-grade managers or professionals.

Comparing and summarising Models 1–3, we find that digital technologies used by workers impact AI use at work more than the education or training required for respondents' occupations or occupational levels. However, the use of digital technologies is likely to be associated with the education or training required for occupations and occupational levels. Hence, a combination of these factors makes AI use in individual workplaces more likely, with digital technologies playing a particularly important role.

The main purpose of the second step of the analysis was to examine the association between autonomy and AI use. The results are presented in Table 4. Therefore, autonomy was used as the dependent variable and AI use, use of digital technologies, education or training required for respondents' occupations, and respondents' occupational levels were used as independent variables.

Model 1 tested for the broad association between autonomy and AI use at work. Without considering other factors, a significant positive association was found ($p < 0.001$). Thus, workers who used AI-based systems reported significantly higher levels of autonomy than those who did not. However, this positive and significant association disappeared when eight digital technologies used at work were included in Model 2. Except for the use of robots and ERP systems, significant positive associations were observed for all digital technologies ($p < 0.001$). This indicates that workers who use digital technologies systematically report higher levels of autonomy. It is possible that the use of these digital technologies increased autonomy, or that such technologies were used primarily where a higher degree of autonomy already existed. Causal relationships cannot be presented using available cross-sectional data in any model. Workers who used robots reported significantly less autonomy ($p < 0.001$), similar to those who used ERP systems ($p < 0.05$). Overall, Model 2 showed that to examine autonomy, it is more important to consider which digital technologies are used at work, rather than merely the use of AI. This was also demonstrated by the significant increase in the r^2 value in Models 1–2.

For a more detailed analysis, in Model 3, digital technologies were removed, and the education or training required for respondents' occupations and occupational levels were included. Again, a significant positive association with AI use was found, albeit the coefficient decreases to roughly half the size of the coefficient in Model 1 ($p < 0.001$). Therefore, without considering digital technologies at work, the significant positive association between autonomy and AI use reappeared, as shown in Model 1. However, both the level of education or training required and the occupational level also showed significant associations with autonomy. Significant positive associations were found between autonomy and higher levels of education or training required for respondents' occupations; thus, people whose jobs required vocational training, a technical college, or a university degree systematically reported more autonomy ($p < 0.001$) than the reference group that required no education or training. Moreover, further associations with the occupational level were found. Compared with the reference group of lower-grade managers or professionals, self-employed

Table 4. Results of the OLS regressions

	(1) Job autonomy	(2) Job autonomy	(3) Job autonomy	(4) Job autonomy
Controls	Yes	Yes	Yes	Yes
Use of AI	0.272***	−0.010	0.093***	−0.021
	(0.014)	(0.014)	(0.014)	(0.014)
Digital technology				
Stationary PC, work station, server		0.049***		0.029***
		(0.004)		(0.004)
Laptop/notebook		0.060***		0.036***
		(0.005)		(0.005)
Smartphone/tablet		0.045***		0.037***
		(0.004)		(0.004)
Robot		−0.046***		−0.034**
		(0.011)		(0.011)
Scanner		0.022***		0.023***
		(0.005)		(0.005)
Digital measurement/diagnostic devices		0.017**		0.009
		(0.006)		(0.006)
Programmes that send/receive messages		0.103***		0.078***
		(0.005)		(0.005)
Programmes that specify the times, places, and resources of work (ERP systems)		−0.011*		−0.005
		(0.005)		(0.005)
Education/training				
(ref. No completed vocational training)			0.000	0.000
			(.)	(.)
Instruction/training			−0.011	0.008
			(0.032)	(0.031)
Completed vocational training			0.263***	0.176***
			(0.019)	(0.019)
Technical college degree			0.295***	0.166***
			(0.027)	(0.027)
University/institution of higher education degree			0.300***	0.158***
			(0.025)	(0.025)
Occupational level				
(ref. Low-grade manager/professional)			0.000	0.000
			(.)	(.)
High-grade manager/professional			0.138***	0.108***
			(0.020)	(0.019)
Routine non-manual employee, higher-grade (in administration/commerce)			−0.132***	−0.052*
			(0.022)	(0.021)
Routine non-manual employee, lower-grade (sales and services)			−0.278***	−0.134***
			(0.025)	(0.025)
Small self-employed with/without employees			0.407***	0.376***
			(0.038)	(0.037)
Skilled manual worker			−0.262***	−0.051
			(0.027)	(0.028)
Semi- and unskilled manual worker			−0.496***	−0.235***
			(0.025)	(0.026)
Agricultural labourer			0.002	0.223***
			(0.060)	(0.059)
Otherwise self-employed			0.409***	0.305***
			(0.037)	(0.036)
Constant	3.442***	2.830***	3.550***	3.033***
	(0.028)	(0.032)	(0.033)	(0.039)

Table 4 (continued)

	(1) Job autonomy	(2) Job autonomy	(3) Job autonomy	(4) Job autonomy
R^2	0.042	0.197	0.178	0.231
N	13,511	13,511	13,511	13,511

Note: Coefficients are shown, and standard errors are in parentheses. The number of cases was fixed for Model 4 to avoid differences owing to missing values. The full results with controls are provided in the appendix

Source: SOEP v37 2020 (EU version); authors' own calculations. For significance levels, * denotes $p < 0.05$; **, $p < 0.01$; and ***, $p < 0.001$

and higher-grade managers or professionals reported significantly more autonomy, whereas unskilled or semi-skilled workers, skilled workers, and routine non-manual employees reported significantly less autonomy.

Model 4 represents the full model with all variables included. The inclusion of additional variables resolves the statistically significant association between autonomy and AI use. Previous associations observed in Model 2 are largely the same for the use of digital technologies in the workplace but are weaker. This also applies to the education required for respondents' occupation and occupational levels.[5] A comparison of Models 2–4 showed that the different variables influenced each other, suggesting that access to digital technologies was also partially associated with the educational requirements for certain occupations and occupational levels. Thus, Model 4 stresses that workers' autonomy is a complex construct and an interplay of different work-related factors associated with autonomy is likely. As a main result, this highlights the inaccuracy of focusing exclusively on the association between autonomy and AI, as often seen in the discourse, and shows that these associations are more complex than theoretically assumed.

5 Discussion and conclusion

This article examines the overall prevalence of AI at work, the determinants that increase the likelihood of AI use, and whether associations between autonomy and AI use can be empirically demonstrated. Given the lack of quantitative empirical evidence in the literature on AI and work, empirical analyses that consider workplace preconditions and characteristics of workers are essential to provide a realistic picture of the current situation in German firms. This is crucial for understanding the paths of AI as a malleable part of a long-term digitalisation of work. The results showed that approximately 38% of all German workers reported using AI-based systems at work, and that AI use was associated with specific digital technologies. Overall, those with a high occupational level were more likely to use AI at work, especially when compared with manual workers. Autonomy at work was found to result from an interplay of different factors rather than as a single effect of AI use, as suggested in the literature.

In detail, the analysis indicates that the current debate on AI and work must be considered differently. While the debate on AI and work has often neglected the complexity of technology diffusion and use (Howcroft and Taylor 2014, 2023) and overemphasised the transformative force of phenomena such as AI (Joyce et al. 2023), the results suggest that AI diffusion may not be inevitable and equal across the working world. Hence, uniform and prerequisite-free diffusion of AI is unlikely. The empirical results of this article highlight that AI use is associated with certain (technical) preconditions in the workplace (MacKenzie and Wajcman 1999; Howcroft and Taylor 2023) and, in this sense, follows path dependency (Hirsch-Kreinsen 2019) rather than being a radical innovation, as discursively expected. Digital technologies in which AI-based systems are embedded are unevenly diffused (Friedrich et al. 2021; Ohlert et al. 2022), which could lead to a specific diffusion pattern of AI-based applications. Therefore, AI-based systems should be understood as part of the general and long-term digitalisation of work because they rely on pre-existing digital technologies.

Consistent with previous literature, AI is currently used for narrow machine-learning pattern-recognition tasks, such as recognising and processing different data sources (Döbel et al. 2018; Konle-Seidl and Danesi 2022), and almost 38% of German workers already use it. While public discourse frequently emphasizes visionary applications of AI, current implementations remain predominantly task-specific. Although this may suggest certain limitations, ongoing advancements and increasing diffusion across work domains suggest that broader impacts on the world of work are plausible and warrant continued observation.

Furthermore, the empirical results also indicate that workers in high occupational levels were more likely to use AI at work, especially in comparison with workers in manual labour. This is in line with research from Denmark (Holm and Lorenz 2022), Germany (Meyer et al. 2022), and a European OECD study (Lane et al. 2023). One possible explanation is that these workers used AI mainly for support and collaboration to obtain more resources for complex tasks while AI systems performed

[5] However, the associations for the use of digital measurement devices, ERP systems, as well as for skilled manual workers were no longer significant. In contrast, respondents in agricultural labour showed a highly significant association.

simple tasks (Goos et al. 2019; Moore 2019; Lane and Saint-Martin 2021). In addition, it seems likely that workers with higher occupational levels have more freedom to decide how often and in what way they want to use AI-based systems (Lane and Saint-Martin 2021). This could increase existing digital inequalities (Lutz 2019; Damelang and Otto 2023) as general access to digital technologies is linked to the occupational level (Friedrich et al. 2021), which also applies to AI-based systems.

Interestingly, a strong positive association was observed between the use of robots and AI use. It is noteworthy that robots are predominantly used in manufacturing industries, which employ a significant proportion of manual workers. This indicates that the prevalence of AI usage among manual workers may be contingent upon the industry in which they are employed. Overall, certain aspects make the use of AI at work more likely, demonstrating that common and simplistic assumptions regarding the ubiquitous diffusion of AI in the working world seem undifferentiated. In other words, the diffusion and use of AI may occur in specific work contexts, particularly where higher-skill, non-routine and non-manual work is needed and specific digital technologies are used.

Regarding job autonomy, the analyses showed a superficial positive association between autonomy and AI use. AI was mainly used in workplaces where workers reported higher levels of autonomy. However, this association disappeared when the use of digital technologies at work was considered, which appeared to be more strongly associated with autonomy. This association was also no longer significant when the use of digital technologies and education required for occupations and occupational levels were added. Thus, these aspects appear to be more important for autonomy than the mere use of AI-based systems, which is the main result of this analysis. Against this background, simplistic assumptions about the impact of AI on job autonomy are questionable. Rather, the findings suggest that the association between autonomy and AI use at work is more nuanced and depends on the interplay between different workplace factors. Moreover, the mere presence of AI at work has limited implications for workers (Shestakofsky 2017). Radical changes in autonomy, often assumed in the discourse, could not be confirmed based on these results. It is plausible that AI is associated with greater autonomy, particularly when it builds upon and extends the capabilities of existing digital technologies, rather than radically breaking with them.

This article has certain limitations. First, as data from a cross-sectional design were utilised for the analyses, only associations between variables could be tested, not causal effects or directions of association. Hence, the results can be interpreted solely as the discovery of cross-sectional associations and not as evidence of change. Therefore, it is necessary to conduct panel studies with repeated survey waves to identify comprehensive changes in AI use and job quality over time.

In addition, the operationalisation of AI is challenging. The survey and methodology used in this article cover only one area of AI, although it is currently the most widely used (Döbel et al. 2018; Konle-Seidl and Danesi 2022). Consequently, the findings and deductions refer to common state-of-the-art machine learning methods for performing certain tasks at work, and thus follow a clear-cut definition of systems that are used as a tool for work. Therefore, systems that could also have an impact on autonomy, such as AI-based algorithmic management systems, are not included in the scope of this study.

The timing of data collection should also be critically considered, as the data were collected prior to the release of popular generative AI applications (such as ChatGPT) in November 2022 (Warning 2023), which likely made AI-based systems and their benefits accessible to a broader range of workers. This might be particularly true for fields in which workers are already familiar with AI-based systems. However, this does not contradict the argument of a long-term digitalization trend, as our analysis has shown that a combination of various factors makes the use of AI in individual workplaces more likely, with digital technologies playing a particularly important role. Overall, it can be assumed that, at present, a larger number of workers report using AI-based systems at work. The data presented here can therefore serve as a valid point of comparison for examining the periods before and after the release of publicly available AI-based systems, such as ChatGPT.

Moreover, many occupations involve tasks that still cannot be performed by AI-based systems, such as those requiring abstract thinking, creativity, and empathy (Moore 2019; Pettersen 2019; Wilkens 2020). Overall, one of the main challenges lies in the fact that the popular discourse on AI and work often refers to AI projects that are insufficiently diffused for inclusion in general household surveys. Further, many of the discussed examples are pilot projects (Burmeister et al. 2019) and rarely used special applications of AI; therefore, they require specific operationalisation. Collecting this information remains a challenge for future research (OECD 2019). It may be necessary to supplement worker surveys with information from IT staff or directly from technical systems to obtain detailed information on different AI technologies.

Finally, our results did not indicate any evidence of changes in tasks resulting from the use of AI at work. However, this topic is widely debated in the context of AI use at work (Dengler and Matthes 2018). The presented findings are based only on the use of current state-of-the-art AI-based systems for performing certain tasks, without providing evidence of an actual change in tasks due

to AI. It is likely that AI-based technology will continue to evolve and that certain technological obstacles can be overcome (Frey and Osborne 2013). This would allow for an expansion in the AI field of application and the substitution of an increasing number of tasks (Dengler and Matthes 2021). Future studies should examine whether AI use leads to the substitution of specific tasks, and thus to a change in the quality of work.

In summary, based on the empirical findings presented here, AI technologies form part of the long-term digitalisation of work. AI use is gradually being integrated into workplaces through embedding within existing digital technologies. The popular notion that AI radically transforms work for everyone is thereby relativised, and we offer a more nuanced picture of the overall prevalence of AI use across workplaces, educational and occupational levels, as well as its impact on job quality.

Abbreviations

AI	Artificial Intelligence
SOEP	(German) Socio-Economic Panel
ERP	Enterprise Resource Planning (System)
SD	Standard Deviation
AME	Average Marginal Effects
OLS	Ordinary Least Squares
VIF	Variance Inflation Factor

Supplementary Information

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Supplementary Material 1

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Data availability
The data that support the findings of this study are available from the DIW Berlin (German Institute for Economic Research) but restrictions apply to their availability; the data were used under license for the current study, and hence, are not publicly available. Socio-Economic Panel (SOEP), data from 1984 to 2020, EU Edition. DOI: https://doi.org/10.5684/soep.core.v37eu).

Declarations

Competing interests
There are no competing interests to declare.

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